

## EDUCATION

### National Yang Ming Chiao Tung University

Bachelor of Science in Applied Mathematics

Hsinchu City, Taiwan

August 2022 – June 2026

### École polytechnique

International Exchange Program

Palaiseau, France

September 2024 – June 2025

## SKILLS

**Programming** Python, C++, and R;  $\text{\LaTeX}$  and **typst**.

**Languages** Mandarin, English (CEFR C1), Japanese (JLPT N2), and French (conversational).

**ASA Courses** Exam P (July 2025), Exam FM (December 2025), VEE Mathematical Statistics, and VEE Accounting & Finance.

## PROJECTS

### Study of Hitting Time and Probability for Random Walks in One and Two Dimensions<sup>1</sup>

July 2025 – February 2026

Research Project, College Student Research Scholarship, National Science and Technology Council

- Derived hitting time distributions and expectations for one-dimensional and two-dimensional simple random walks.
- Established an  $N^2$  scaling for expected hitting times in bounded domains and linked discrete probability coefficients to torsional rigidity in beam theory, providing a cross-disciplinary analytical framework for evaluating boundary behaviors in random walks.

### The Gambler's Ruin Problem for One-Dimensional Random Walks: Simple Symmetric Random Walk and Some Extensions<sup>1</sup>

September 2025 – December 2025

Semester Project, [Individual Directed Study \(I\)](#), National Yang Ming Chiao Tung University

- Verified the solution  $x/r$  for simple symmetric walks and extended the scaling  $\asymp (x+1)/r$  to finite-step-length and finite-variance models, adapting classical stochastic models to broader conditions.
- Awarded **Best Project** (Departmental) and **Best Presentation** (2025 Inter-University Conference).

### Credit Card Default Detection Using Machine Learning

February 2025 – June 2025

Final Group Project, [Machine Learning](#), École polytechnique

- Engineered predictive pipelines in Python (`scikit-learn`) to address imbalanced datasets via optimizing the F1-score to capture rare default events.
- Evaluated the trade-off between interpretability and accuracy by comparing baseline logistic regression against ensemble methods like XGBoost, demonstrating that transparent models can achieve comparable performance.

### Chessbot Project: Cagnus

November 2024 – January 2025

Final Group Project, [Object-oriented Programming](#), École polytechnique

- Implemented bitboard representations and minimax evaluation in pure C++.
- Placed 9th of 26 in the class competitive tournament.

### Before Finance<sup>1</sup>

June 2024 – July 2024

[Summer Research](#), Department of Applied Mathematics, National Yang Ming Chiao Tung University

- Analyzed hitting times for simple random walks in *hitting time of 1* and the *gambler's ruin problem*.
- Constructed measure-theoretic proofs for martingale properties, including the optional sampling theorem, establishing a theoretical foundation for stochastic analysis and quantitative modeling.

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<sup>1</sup>Supervised by Prof. [Yuki CHINO](#).

**WORK  
EXPERIENCE**

**Teaching Assistant** September 2025 – June 2026 & September 2023 – June 2024  
Department of Applied Mathematics, National Yang Ming Chiao Tung University  
AY 2025 [Probability Theory](#) and [Calculus A \(II\)](#) under Prof. [FANG Xiang](#).  
AY 2023 [Calculus A \(I\)](#) and [Calculus A \(II\)](#) under Prof. Yuki CHINO.

**Private Tutor** March 2026 – June 2026, September 2024 – June 2025, & April 2023 – June 2023  
Tutored students with customized handouts.

**Academic Tutor** October 2023 – November 2023 & April 2024 – May 2024  
Center of Teaching and Learning Development, National Yang Ming Chiao Tung University  
Tutored overseas students in Calculus A and conducted weekly sessions with customized [handouts](#).

**HONORS**

**Mathematics Presidential Award** March 2026, September 2023, & February 2023  
Department of Applied Mathematics, National Yang Ming Chiao Tung University  
For my academic performance in applied mathematics in my undergraduate study.

**College Student Research Scholarship** July 2025 – February 2026  
[National Science and Technology Council](#)  
For the project “Study of Hitting Time and Probability for Random Walks in One and Two Dimensions”  
under research grant 114-2813-C-A49-139-M.

**[PTSGI.com 23rd Translation Scholarship](#)** January 2026  
[PTSGI.com](#) (Presidential Translation Service Group International)  
Selected as one of the 30 recipients in the translation competition for Japanese-to-English translation.

**College of Science Dean Award** October 2025  
College of Science, National Yang Ming Chiao Tung University  
For ranking in the top 10% of the department in my first three years of undergraduate study.

**Outstanding Student Overseas Exchange Scholarship** June 2024  
National Yang Ming Chiao Tung University & the Ministry of Education  
For my exchange program at École polytechnique in France.

**Honor Student** May 2023  
National Yang Ming Chiao Tung University  
For ranking in the top 5% of the department in my first semester of undergraduate study.